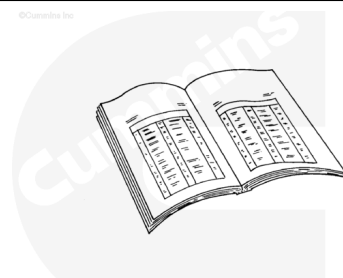


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.



ck800wa

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. Refer to the OEM service manual.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>)
- Remove the oil lubricating pan adapter covers. Refer to Procedure 007-026 In Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html>)
- Remove the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)

- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)

- Remove the fuel filters, if needed. Refer to Procedure 006-015 in Section 6.

(/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)

- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)

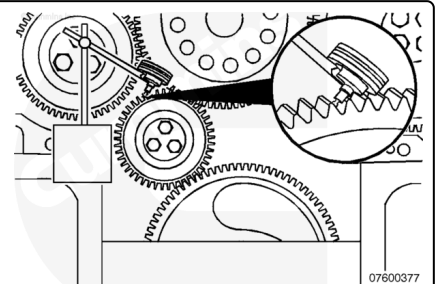
- Remove the lubricating oil transfer tube. Refer to Procedure 007-040 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)

Remove

The procedures for removing the lubricating oil pump on the QSK45 and QSK60 engines are similar. The oil pump for the QSK60 is shown.

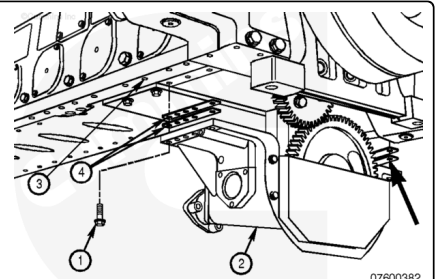
It is recommended that the oil pump to crankshaft gear backlash be measured before removing the pump to check for a worn gear.



Lubricating Oil Pump Idler Gear Backlash		
mm		in
0.21	MIN	0.008
0.39	MAX	0.015

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Loosen all of the mounting capscrews (1).

Support the lubricating oil pump.

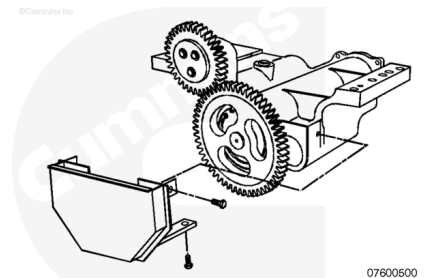
Remove the mounting capscrews (1).

Gently lower the lubricating oil pump (2) away from the engine (3). Note the amount and location of any shims (4).

Disassemble

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



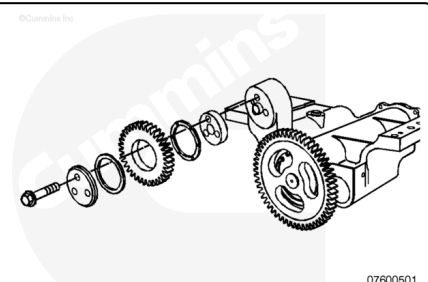
⚠ CAUTION ⚠

The lubricating oil pump must be properly supported to prevent falling and damaging the pump.

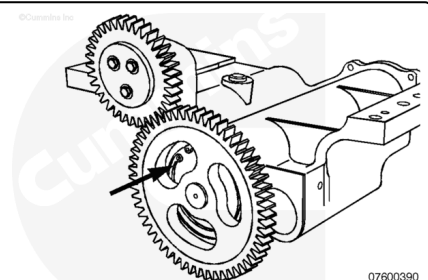
Remove the four gear cover mounting capscrews and remove the gear cover from the lubricating oil pump.

Remove the three capscrews securing the idler gear and shaft to the lubricating oil pump.

Remove the gear retainer, thrust bearing, gear, and shaft.



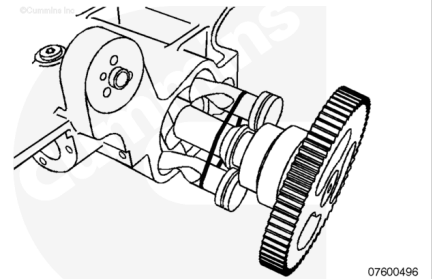
Remove the main rotor bearing retaining snap ring.



Slide the rotor assembly forward 152 to 203 mm [6 to 8 in].

To reduce the possibility of damage to the rotor assembly, tie a lint free rag or wire tie around all three rotors.

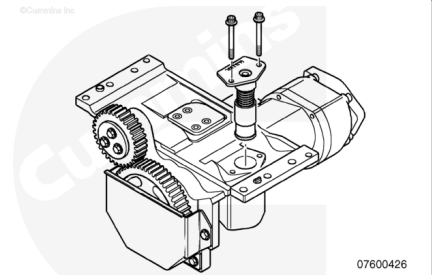
Remove the rotor assembly from the lubricating oil pump housing.



Remove the lubricating high pressure relief valve.

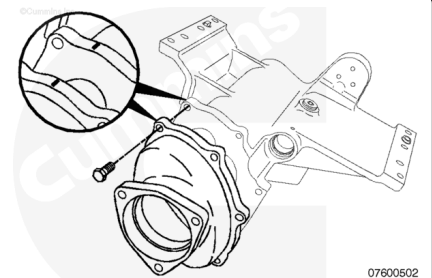
Refer to Procedure 007-021 in Section 7.

(</qs3/pubsys2/xml/en/procedures/56/56-007-021-tr.html>)



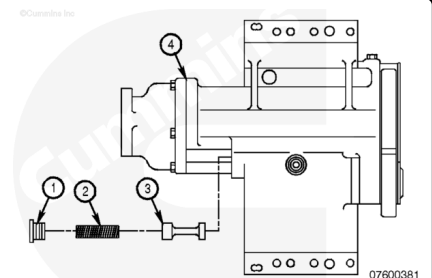
Mark the inlet head and the lubricating oil pump body for alignment during assembly.

Remove the inlet head mounting capscrews and the inlet head.



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear the appropriate eye and face protection when removing the straight threaded o-ring plug holding in the oil pressure regulator. The spring is under pressure and can cause personal injury.

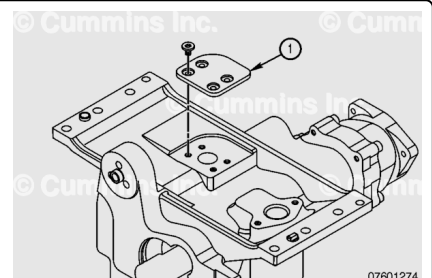


Remove the lubricating oil pressure regulator. Refer to Procedure 007-029 in Section 7.

(</qs3/pubsys2/xml/en/procedures/56/56-007-029.html>)

Remove the four capscrews holding down the oil pressure regulator cover plate (1).

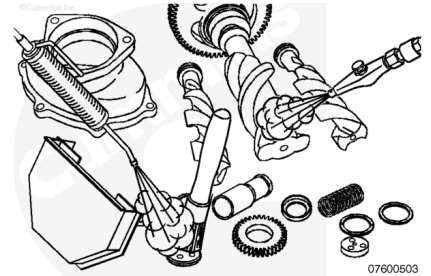
Remove the cover plate.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

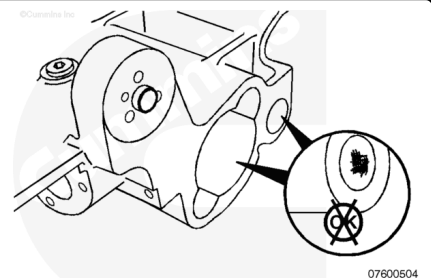
Clean the lubricating oil pump components with solvent. Use a bristle brush to clean the oil drilling in the idler gear shaft.

Use a steam cleaner to rinse the components and dry with compressed air.

Inspect the following areas of the lubricating oil pump for damage or blockage.

- Rotor bore housing
- Inlet and outlet ports
- Internal oil passages
- Lubricating oil pump housing
- Inlet head
- Pressure regulator bore.

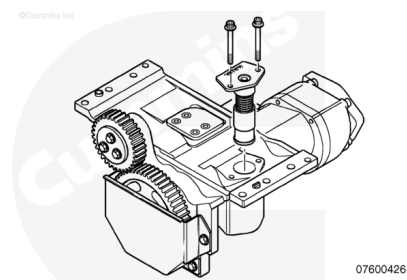
If the lubricating oil pump housing is damaged, it **must** be replaced.



Inspect the lubricating oil high pressure relief valve assembly and housing for damage.

Refer to Procedure 007-021 in Section 7.

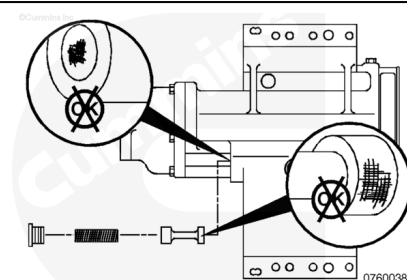
(/qs3/pubsys2/xml/en/procedures/56/56-007-021-tr.html)



Inspect the lubricating oil pressure regulator assembly and housing for damage.

Refer to Procedure 007-029 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-029.html)



There is a step in the power rotor by design. Do **not** mistake this step for abnormal wear.

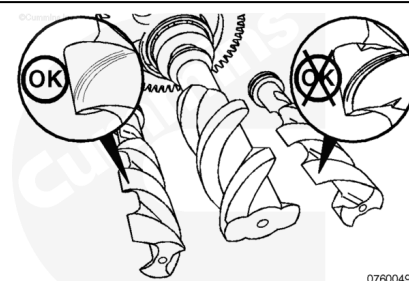
Inspect the three rotors for any damage or abnormal wear.

Minor damage can be removed with a hand stone.

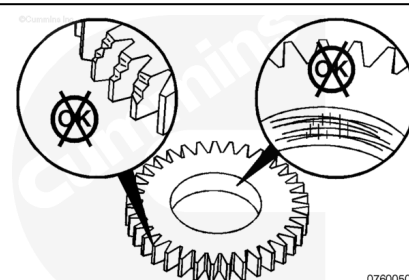
Inspect the main rotor bearing for damage or restricted rotation.

Inspect the bearing retainer snap-ring for bending or twisting.

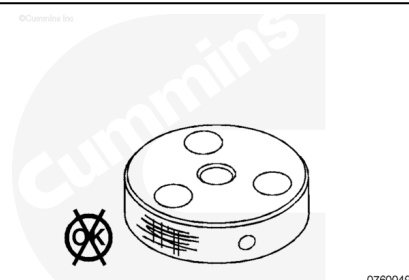
Replace damaged components.



Inspect the idler gear and bushing for wear, signs of seizure, and splitting.

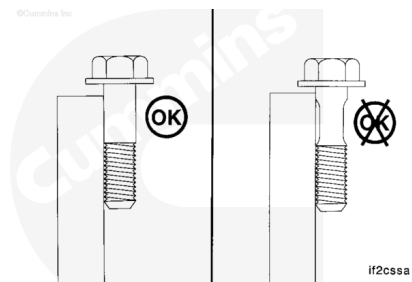


Inspect the idler shaft for wear, signs of seizure, and other damage.

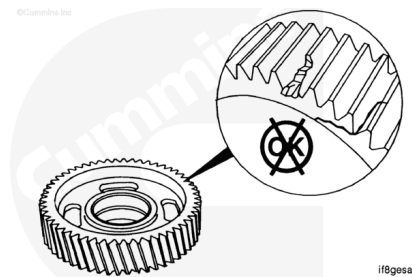


Use a straight edge to inspect the retaining capscrews for necking. If necking is observed, do **not** reuse the capscrews.

The capscrews **must** be replaced.



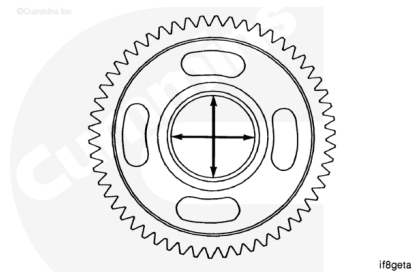
Inspect the idler gear for chipped, broken, or cracked teeth.



Measure the inside diameter of the idler gear bushing bore.

Lubricating Oil Pump Idler Gear Bushing Bore Inside Diameter

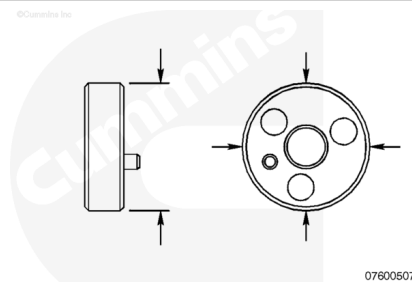
mm		in
60.045	MIN	2.364
60.100	MAX	2.366



Measure the outside diameter of the idler shaft.

Lubricating Oil Pump Idler Gear Shaft Outside Diameter

mm		in
59.975	MIN	2.361
60.006	MAX	2.363

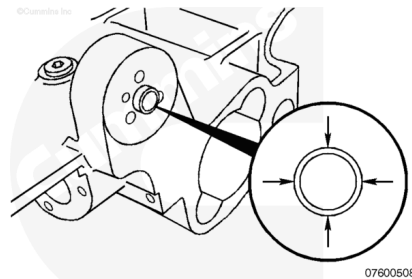


Inspect the ring dowel outside diameter for damage. If damaged, the dowel **must** be replaced.

Measure the outside diameter of the ring dowel.

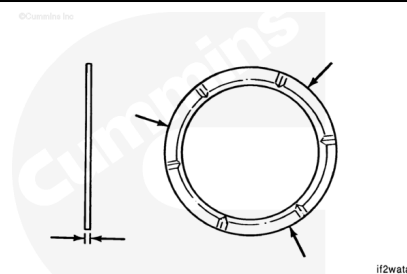
Lubricating Oil Pump Ring Dowel Outside Diameter

mm		in
19.217	MIN	0.757
19.243	MAX	0.758

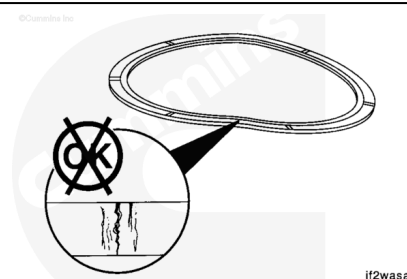


Measure the thrust washer thickness in three places, 120 degrees apart.

Lubricating Oil Pump Idler Thrust Washer Thickness		
mm		in
2.47	MIN	0.097
2.53	MAX	0.099



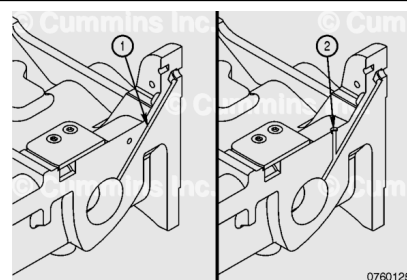
Flex the thrust washers approximately 3 to 6 mm [1/8 to 1/4 in] and inspect the surfaces for cracks.



Machine

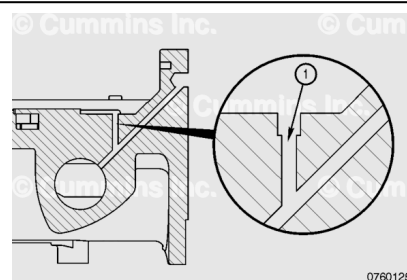
A product improvement to the lubricating oil pump allows the lubricating idler bushing to be fed with filtered (clean) oil.

This illustration shows the original non-filtered (1) and new filtered (2) oil flow through the pump to the idler bushing.



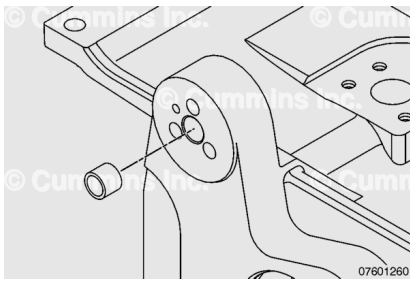
The following steps describe how to convert a lubricating oil pump from an old to a new style design.

A new oil rifle drilling is required as shown (1).



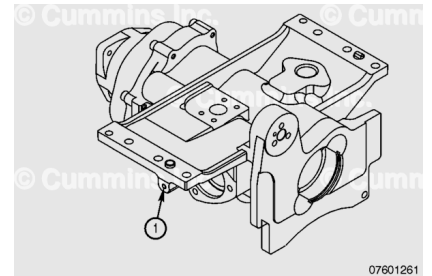
Disassemble the pump as described in the disassemble section.

Remove the idler gear shaft dowel. Use a dowel pin puller, Cummins® Service Tool, Part Number 3163720, or equivalent.



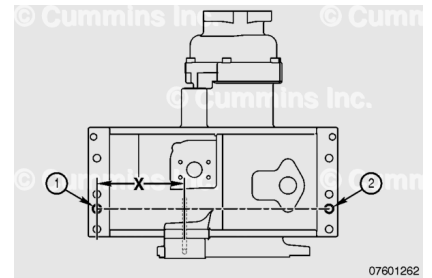
Remove the existing oil rifle AVSEAL® (1).

Use an AVSEAL® removal kit, Cummins® Service Tool, Part Number 3823154 or equivalent.



With the housing secured, drill the new oil rifle at the distance (X) measured from the center of the lower dowel hole (1) on the plane running between the center of the 2 dowel holes (1 and 2).

Note : The new oil rifle position is the same for QSK45 and QSK60 lubricating oil pumps.



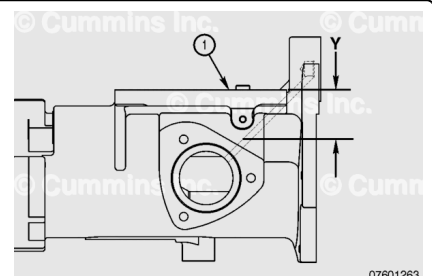
Hole Diameter		
mm		in
6.94	MIN	0.273
7.34	MAX	0.289

X - Hole Depth		
mm		in
163.14	MIN	6.423
163.54	MAX	6.439

The new oil rifle **must** be drilled to the depth shown (Y).

Note : The drill depth is measured from the base of the lubricating oil pump where the dowel is located (1).

The new oil rifle depth is the same for QSK45 and QSK60 lubricating oil pumps.

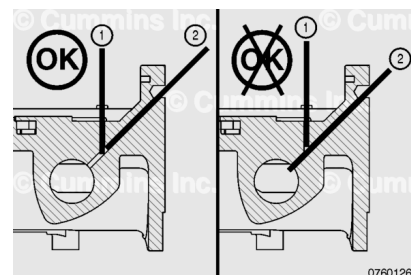


Dimension Y

mm		in
62.2	MIN	2.449
64.2	MAX	2.528

Be sure there is full break through to the existing oil line after drilling.

To check for full break through, use two 6mm [0.236 in] diameter rods, 70 mm [2.75 in] (1) and 200 mm [7.87 in] (2) long placed in the oil holes. Insert rod 1 into the new oil hole. Insert rod 2 into the existing oil hole. If rod 2 can **not** pass all the way through, the existing oil hole full breakthrough has been achieved (A). If rod 2 can pass all the way through the existing oil hole, full breakthrough has **not** been achieved (B).



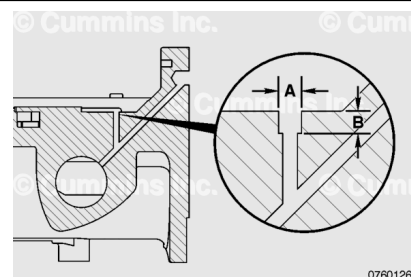
With the pump still secured in position, ream out the new oil hole to the following diameter (A) and depth (B) from the housing surface as shown.

Hole Diameter (A)

mm		in
7.8	MIN	0.307
8.4	MAX	0.331

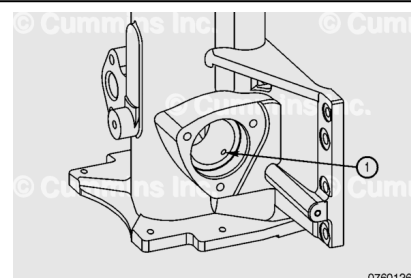
Hole Depth (B)

mm		in
8.5	MIN	0.335
9.0	MAX	0.354



⚠ WARNING ⚠

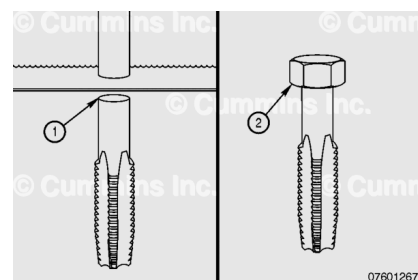
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Stand the lubricating oil pump housing on the front face. This will allow access to the internal oil drilling (1).

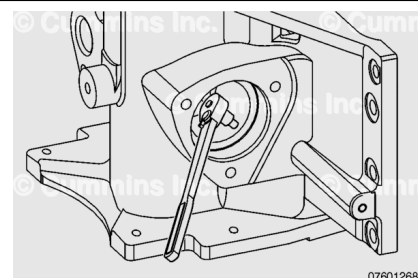
The old oil rifle **must** be plugged, use a M8 x 8mm long pipe plug.

Use a tap of the same size and pitch to tap the holes. If the tap is too long it **must** be trimmed down (1) and a nut pressed on or welded to the end (2) to allow the tap to be turned with a wrench. The new tap length shown in the illustration is approximately 40 mm [1.57 inch]. Pressing the nut on is preferred, as welding can affect tap strength.

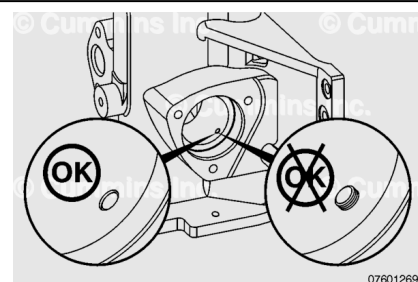


Tap the internal oil rifle hole.

Note : Tap the hole deep enough so that the pipe plug will not protrude into the bore once installed.

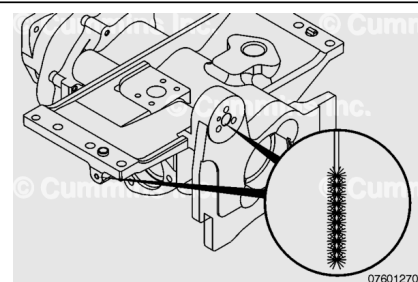


Once the hole is tapped, temporarily install the plug into the hole to make sure the plug does **not** protrude into the bore. If the hole is tapped to shallow, remove the plug, tap the hole deeper and install the plug to check the installation depth. If the hole is deep enough, remove the plug and continue.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Make sure all oil passages are thoroughly cleaned, otherwise engine damage will occur.

Use solvent and a nylon brush to thoroughly clean the lubricating oil pump housing and all internal drillings. The internal drillings can be cleaned using a passage cleaning brush, Cummins® Service Tool, Part Number ST 876.

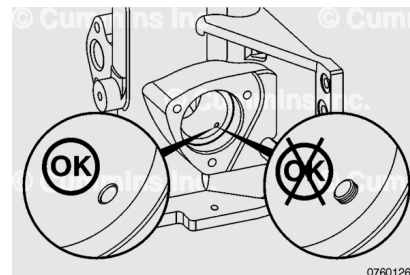
Dry with compressed air.

Apply Loctite™ 271 (Cummins® Part Number 3375068) or equivalent to the pipe plug threads and install.

Tighten the pipe plug until it bottoms in the threaded hole.

Torque Value: 25 n•m [221 in-lb]

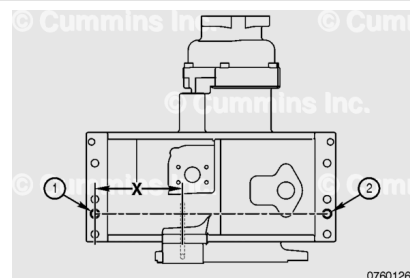
Make sure the pipe plug does **not** protrude into the bore once installed.



Insert a 6mm [0.236 in] diameter rod, 200 mm [7.87 in] long in to the oil rifle at the side of the pump (1).

The rod **must** enter the oil hole a minimum of 185 mm [7.28 in].

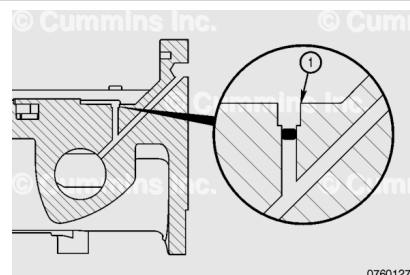
This is to make sure that when an AVSEAL® is installed into the new external oil rifle drilling, the oil rifle is **not** blocked.



Install an AVSEAL®, Part Number 3640775, into the new external oil rifle drilling (1). Use Service Tool, Part Number 3824154.

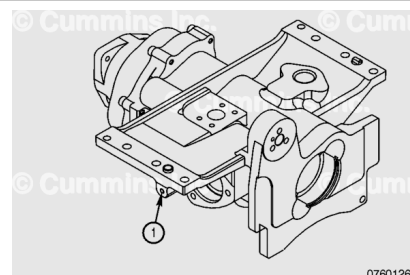
Note : The AVSEAL® **must not** protrude out of the casting.

Remove the rod that was inserted into the side oil rifle in the previous step.



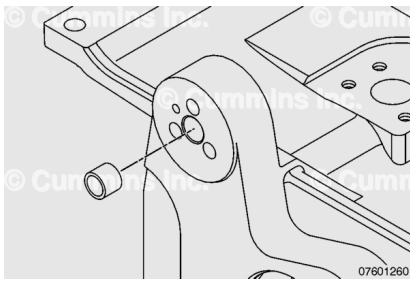
Replace the AVSEAL® removed earlier from the oil rifle at the side of the pump (1).

Use AVSEAL®, Part Number 3640775, and Service Tool, Part Number 3824154.



Use a soft mallet to install the idler gear shaft dowel by tapping it into place.

Assemble the lubricating oil pump.

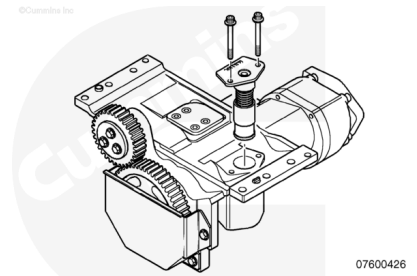


Assemble

Install the lubricating oil high pressure relief valve.

Refer to Procedure 007-021 in Section 7.

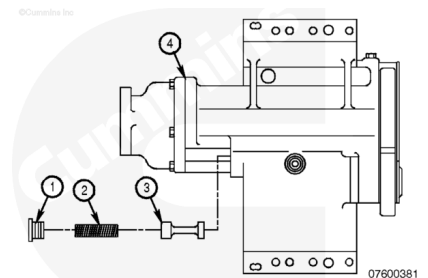
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Install the lubricating oil pressure regulator.

Refer to Procedure 007-029 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-029.html)



For assembly purposes, the recommendation is to have the lubricating oil pump idler gear part number facing away from the cylinder block.

Lubricate the idler gear bushing and thrust bearing with Lubriplate™ 105 or equivalent.

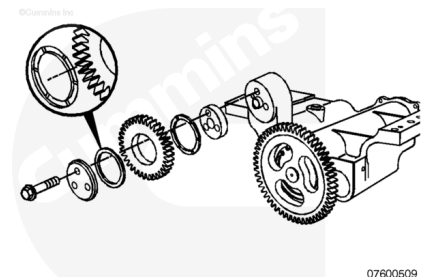
Lubricate the capscrews with clean engine oil.

Place the idler gear shaft on the ring dowel.

Verify the oil drilling in the idler shaft is in the 6 o'clock position.

Place the thrust bearing on the gear with the grooved side of the thrust bearings facing the gear.

Place the idler gear and thrust bearings onto the idler gear shaft.

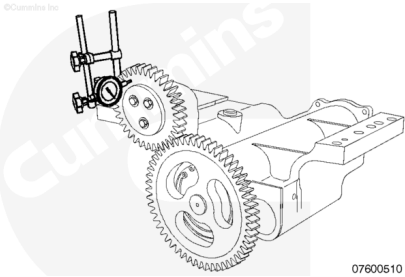


Install the idler gear retainer and capscrews.

Tighten the capscrews.

Torque Value: 55 n•m [41 ft-lb]

Use magnetic base, Part Number 3377399, and dial indicator, Part Number 3376050, to measure the idler gear end clearance.



Lubricating Oil Pump Idler Gear End Clearance

mm		in
0.080	MIN	0.003
0.360	MAX	0.014

If the end clearance is **not** within specifications, check for foreign material between the components and for proper installation of the thrust bearings.

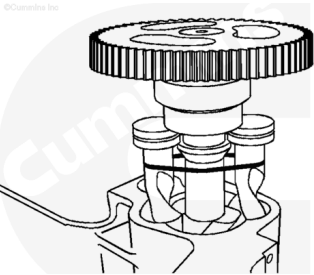
⚠ CAUTION ⚠

Do not allow the rotors to drop into the bore. Damage to the rotors or housing can result.

Lubricate the rotors and lubricating oil pump housing with clean engine oil.

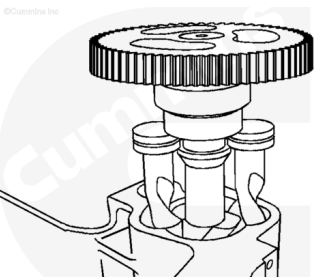
Set the lubricating oil pump housing upright so the rotor bores are vertical and the open end is facing up.

To avoid damage and aid in the installation of the rotors, tie a lint free rag or wire tie around all three rotors.



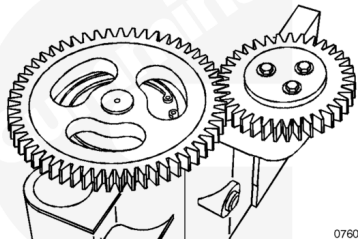
Slide the rotor assembly into the housing bore to a point where the wire tie or lint free rag can be removed.

Remove the wire tie or lint free rag.



Continue installing the rotor assembly into the pump housing.

Use care when aligning the power rotor gear with the idler gear to avoid damage to the gears.



07600513

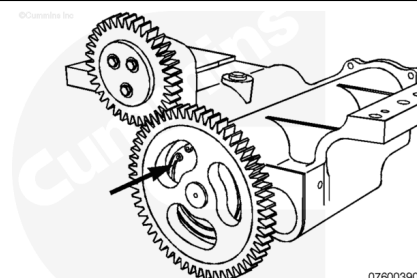
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Set the pump on its side and install the snap ring into the snap ring groove.

Rotate the pump gears to check for binding.

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07600390

Install the pump cover and four mounting capscrews.

Tighten the capscrews.

Torque Value:

M8 23 n·m [163 in-lb]

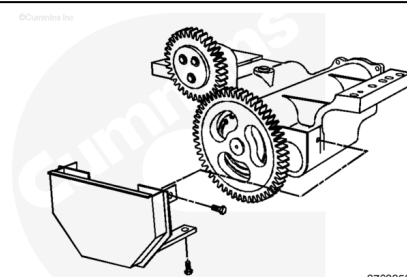
Torque Value:

M10 45 n·m [33 ft-lb]

Torque Value:

M12 80 n·m [59 ft-lb]

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07600500

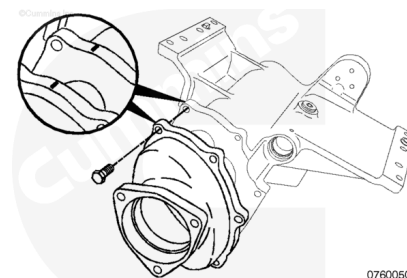
Install a new gasket and the suction head to the lubricating oil pump body, aligning the mark made during assembly.

Apply Loctite™ 271 (Cummins® Part Number 3375068) or equivalent to the capscrew threads.

Install and tighten the suction head capscrews.

Torque Value: 54 n·m [40 ft-lb]

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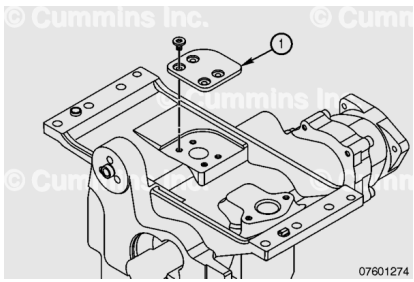
07600502

Install the oil pressure regulator cover plate (1).

Apply Loctite™ 271 (Cummins® Part Number 3375068) or equivalent to the capscrew threads.

Install and tighten the cover plate capscrews.

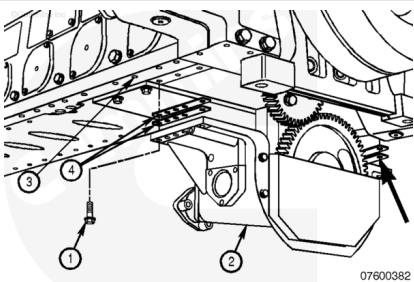
Torque Value: 20 n•m [177 in-lb]



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the pump (2) and shims (4) on the engine (3).

Install the mounting capscrews (1).

Tighten the capscrews.

Torque Value:

M12

1. 80 n•m [59 ft-lb]

Torque Value:

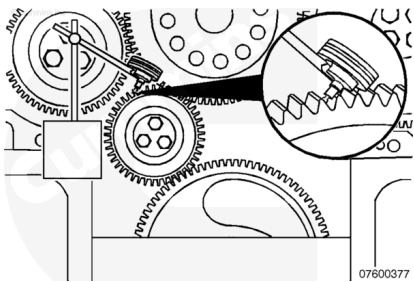
M10

1. 45 n•m [33 ft-lb]

⚠ CAUTION ⚠

Backlash measurement is critical. If not properly adjusted, lubricating oil pump or engine failure will occur.

It is required to measure the lubricating oil pump backlash three times. It **must** be measured on the inverted engine when the pump is mounted, after the pump plumbing is installed, and again after the engine is rotated upright.



Lubricating Oil Pump Idler Gear Backlash

mm		in
0.21	MIN	0.008

Lubricating Oil Pump Idler Gear Backlash

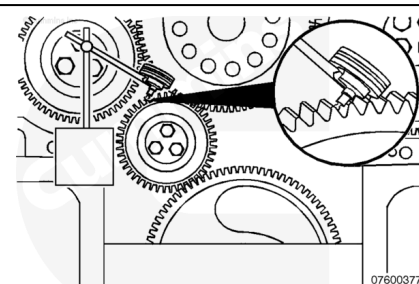
mm		in
0.39	MAX	0.015

If the backlash is **not** within specification, adjust the lubricating oil pump.

Adjust

⚠ CAUTION ⚠

Backlash measurement is critical. If not properly adjusted, lubricating oil pump or engine failure will occur.



It is required to measure the lubricating oil pump backlash three times. It **must** be measured on the inverted engine when the pump is mounted, after the pump plumbing is installed, and again after the engine is rotated upright.

Use a dial indicator to measure the lubricating oil pump idler gear to crankshaft gear backlash.

Backlash - Lubricating Oil Pump Idler Gear to Crankshaft Gear

mm		in
0.21	MIN	0.008
0.39	MAX	0.015

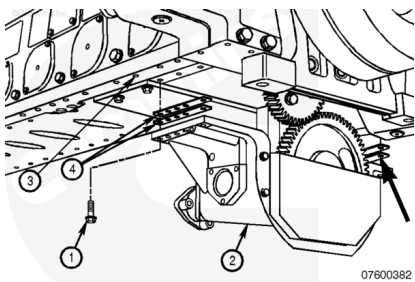
If the backlash is less than specifications, remove the pump and install shims between the pump and the engine. Shims are available 0.13 mm [0.005 in], Part Number 4015782, and 0.25 mm [0.010 in], Part Number 4015783.

If the backlash is greater than specifications, remove any shims, and measure the backlash again. If no shims were used, the pump drive gear or the crankshaft gear is worn and **must** be replaced.

An equal amount of shim thickness **must** be used under both mounting feet, if shims are required.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Position the shims (4) between the pump (2) and the engine block (3).

Install the pump (2) and shims (4) on the engine with the six capscrews (1).

Tighten the capscrews.

Torque Value:

M12

1. 80 n•m [59 ft-lb]

Torque Value:

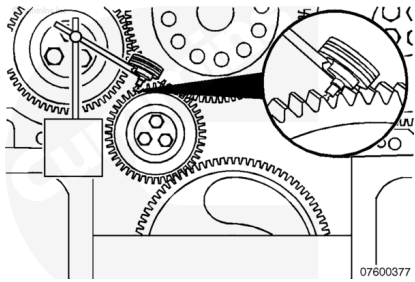
M10

1. 45 n•m [33 ft-lb]

Check the lubricating oil pump backlash.

Lubricating Oil Pump Idler Gear to Crankshaft Gear Backlash

mm		in
0.21	MIN	0.008
0.39	MAX	0.015



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)

- Install the lubricating oil transfer tube (2). Refer to Procedure 007-040 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)
- Install the oil pan adapter (2). Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Install the lubricating oil pan adapter covers. Refer to Procedure 007-026 In Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html)
- Install the new lubricating oil filters. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)
- Install the new fuel filters, if removed. Refer to Procedure 006-015 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Start the engine and check for proper operation.